

1. 256 possible 8-bit instructions exist in this system format.
2. 01000011 (010)(0)(00)(11): This example would be considered invalid due to the instruction (010), as it calls LOAD; therefore, memory access needs to be on (1) but instead is off (0).
3. Both s1 and s2 are ignored during the OUT instruction, and this could cause a practical issue as the values are still there visually. Unlike the system, a human can't just "visually ignore" sections when reading/debugging.
4. (Assuming register 1 is empty and the rest are full).
00000100: This reads register 1, I would confirm it's empty.
01110100: This stores register 1 to register 0.
01110110: This stores register 1 to register 2.
01110111: This stores register 1 to register 3.
Now all registers should be full
5. Q1. Reads input into register 2
Q2. 01011101
Q3. Add registers 1 and 2, storing results in register 1
Q4. 00100000
6. Register 1 holds the final sum, and it's important because register 1 is used for outputs
7. Before adding the numbers, both registers 1 and 3 contained variable x
8. Binary values can be interpreted as the number seen in registers or columns 5-8. In instruction code or column 1-3, binary values are instructions.
9. Q1. 01011101 Load from register 3 to 1
Q2. 11000110 Bitwise AND of registers 1 and 2
Q3. 01110111 Store register 1 to 3
Q4. 10100101 Subtract register 1 from 1, and store in 1
10. The program takes the variable x and checks it with 1 stored in register 2. If x is odd 1; if even 2.
11. The last step clears register 1 for future use
12. Q1. 11100010
Q2. E2
Q3. Bitwise OR of registers 0 and 2
Q4. 10100001
Q5. 161
Q6. Subtract registers 0 and 1
Q7. 01110111
Q8. 119
Q9. 77

13. It is getting the bitwise OR of register 0 and 2, then subtracting register 1 from it and replacing x with the result.

14. Q1. 10100000

Q2. A0

Q3. Subtract register 0 and 0

Q4. 10100101

Q5. 165

Q6. Subtract register 1 and 1

Q7. 170

Q8. AA

Q9. Subtract register 2 and 2

Q10. 10101111

Q11. AF

Q12. Subtract register 3 and 3

15. This program clears all registers

16. This situation would be helpful in any case where the program needs to be cleared or reset.